

Cyberdeck Build — Cheat Sheet

Last-minute recall reference. For the reasoning behind each row, see [LESSON.pdf](#) or the interactive tour.

CONCEPT	RULE
Runtime estimate	battery mAh × efficiency ÷ total load mA ≈ hours
Power bank efficiency	~80-85% usable — cells are 3.7V, boosted to 5V with real conversion loss
Measure, don't guess	Use a USB power meter under real workload, not datasheet peak/idle figures
DSI vs HDMI touch	DSI = one cable, best drivers. HDMI touch = video cable + separate USB touch cable
Display rotation	Rotating video ≠ rotating touch — touch needs its own transform matrix
Build order rule	Electronics fully working loose on a bench, always before any enclosure work
Wiring	Route every wire with slack (a service loop) — taut wiring fails under flexing over time
Thermal throttling	The Pi downclocks itself when hot — looks like a slow machine, not a broken one
Heatsink rule	Only works with airflow — a heatsink in a sealed box is just extra thermal mass
Battery safety	Never use bare LiPo/Li-ion cells without a BMS (protection circuit)
Calculated vs measured runtime	Expect measured to run somewhat shorter — treat calculated as a floor, not a promise
OS trimming	Raspberry Pi OS Lite + minimal window manager saves real power/thermal headroom